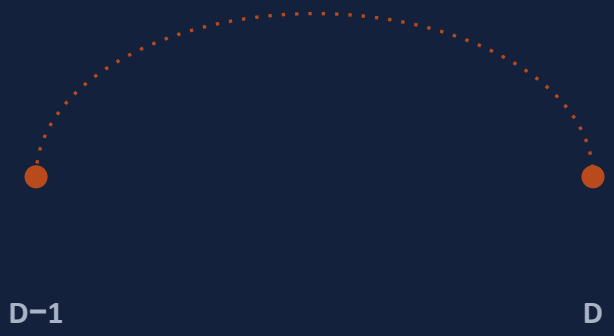
AEROCAST · ADVANCED MACHINE LEARNING · PHASE 1

Flight Delay Prediction via Gradient Boosting and Temporal Deep Learning

Using only what is known by the end of the previous day, how well can we predict which New York departures will arrive 15 or more minutes late — and can we trust the probabilities?



336,776 flights · EWR · JFK · LGA · January–December 2013 · decision time: end of day D−1

Every evening, planners need tomorrow's delay risk — before a single flight departs

DAY D−1

DAY D

Known: tomorrow's published schedule and every earlier flight whose outcome has been reported

23:59 Decision
probability for every flight on D

Unknown: how any flight on day D actually goes — that is what we predict

TARGET

Arrival delay ≥ 15 minutes

is_delayed = 1, else 0 (BTS definition)

UNIT

One scheduled flight

a binary classification per flight

OUTPUT

A calibrated probability

turned into a flag only if a usable threshold exists

Target definition: U.S. Bureau of Transportation Statistics on-time convention [2]. Decision time: D-012.

One year of New York departures: about one flight in four arrives 15+ minutes late

336,776

flights scheduled from EWR, JFK and LGA in 2013, 21 columns

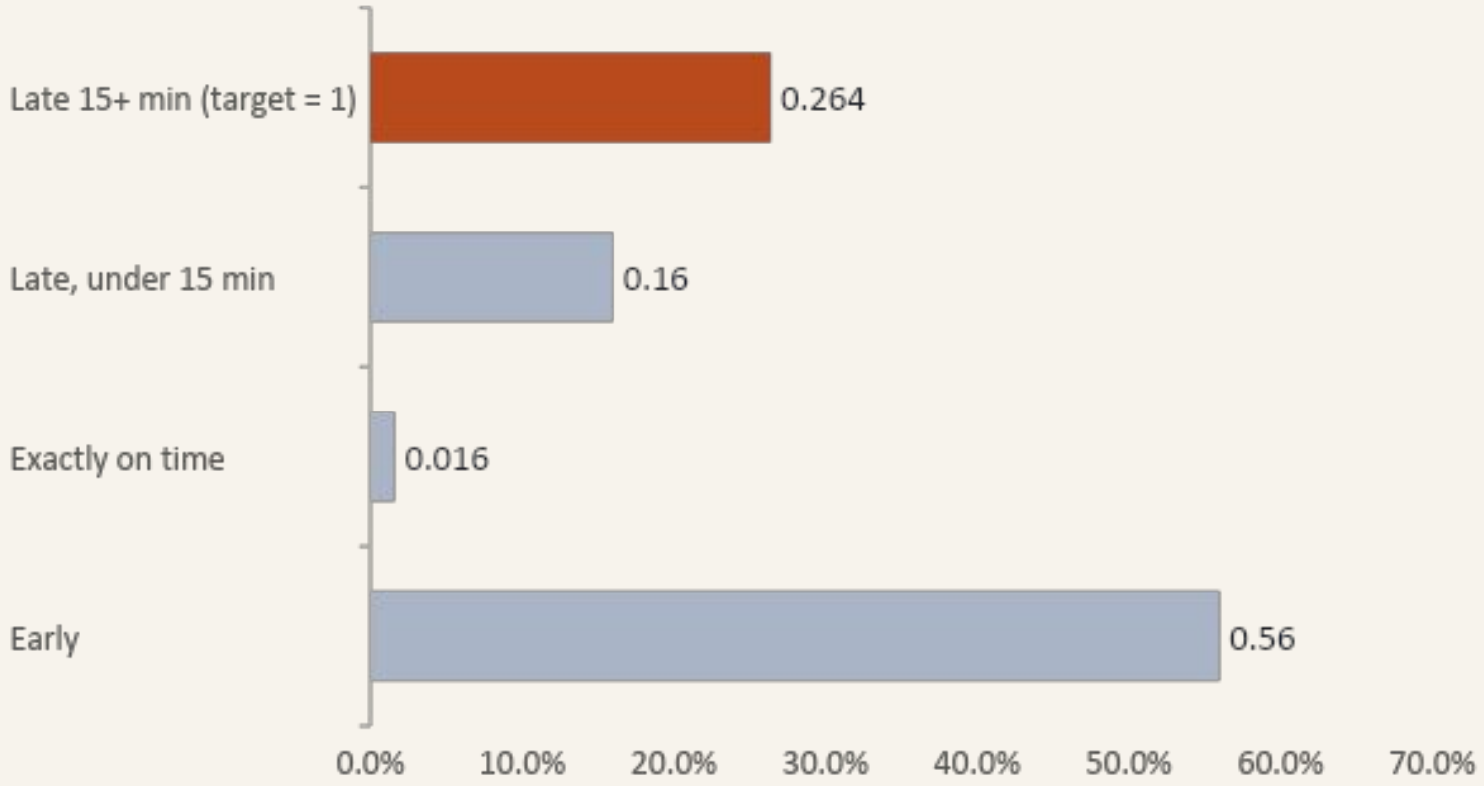
327,346

with an observed arrival delay; the 9,430 cancelled or without an arrival record have no label and are excluded

0

duplicate rows or impossible values; departure delay and tail number leak the outcome, so they are never features

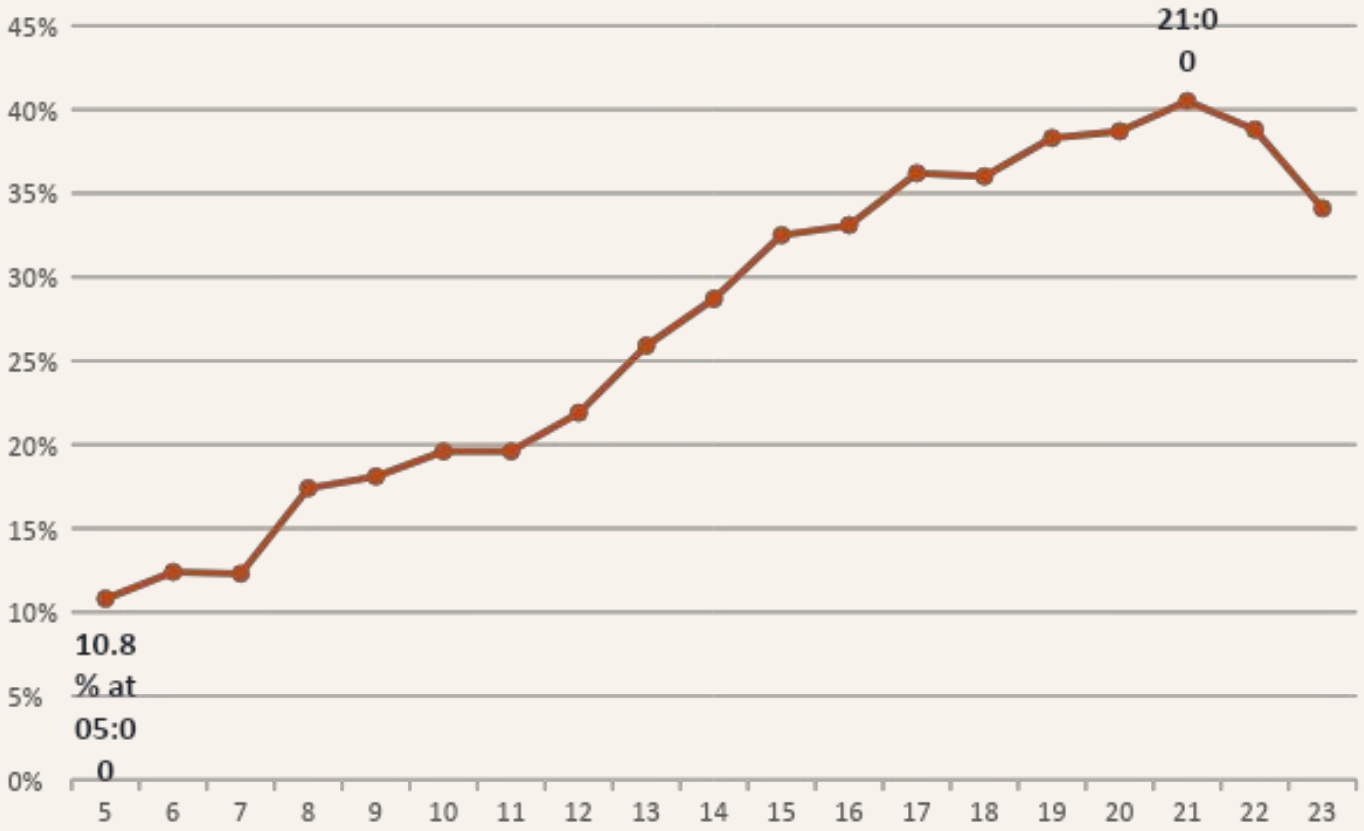
Arrival relative to schedule, training period (217,727 flights)



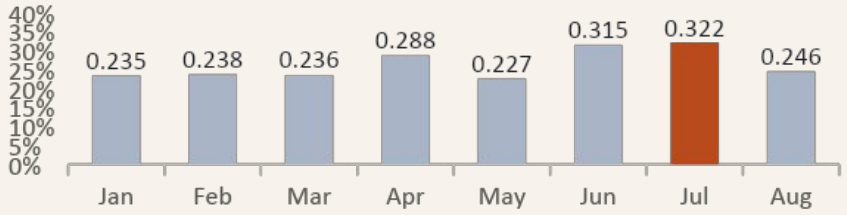
Data: Kaggle Flights dataset [1]; on-time definition [2]. Cleaning: notebook 01 (Jan–Oct); outcome shares: notebook 02 (training period, Jan–Aug).

Delay risk builds through the day — time of day is the clearest single signal

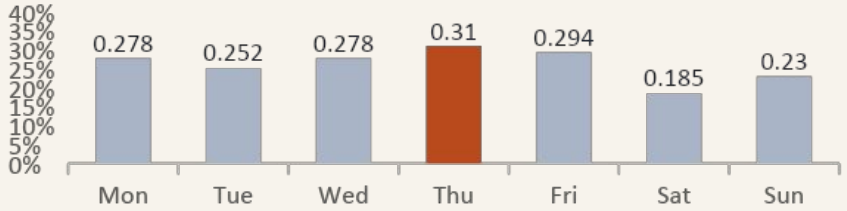
Share of flights delayed 15+ min, by scheduled departure hour



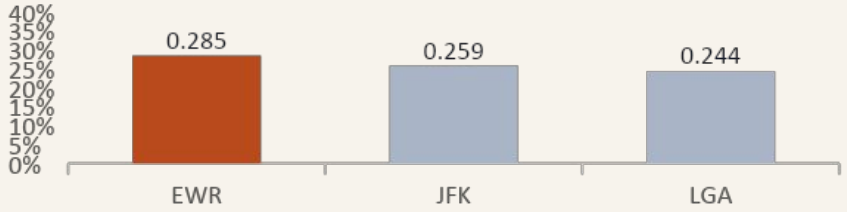
By month (Jan–Aug)



By weekday



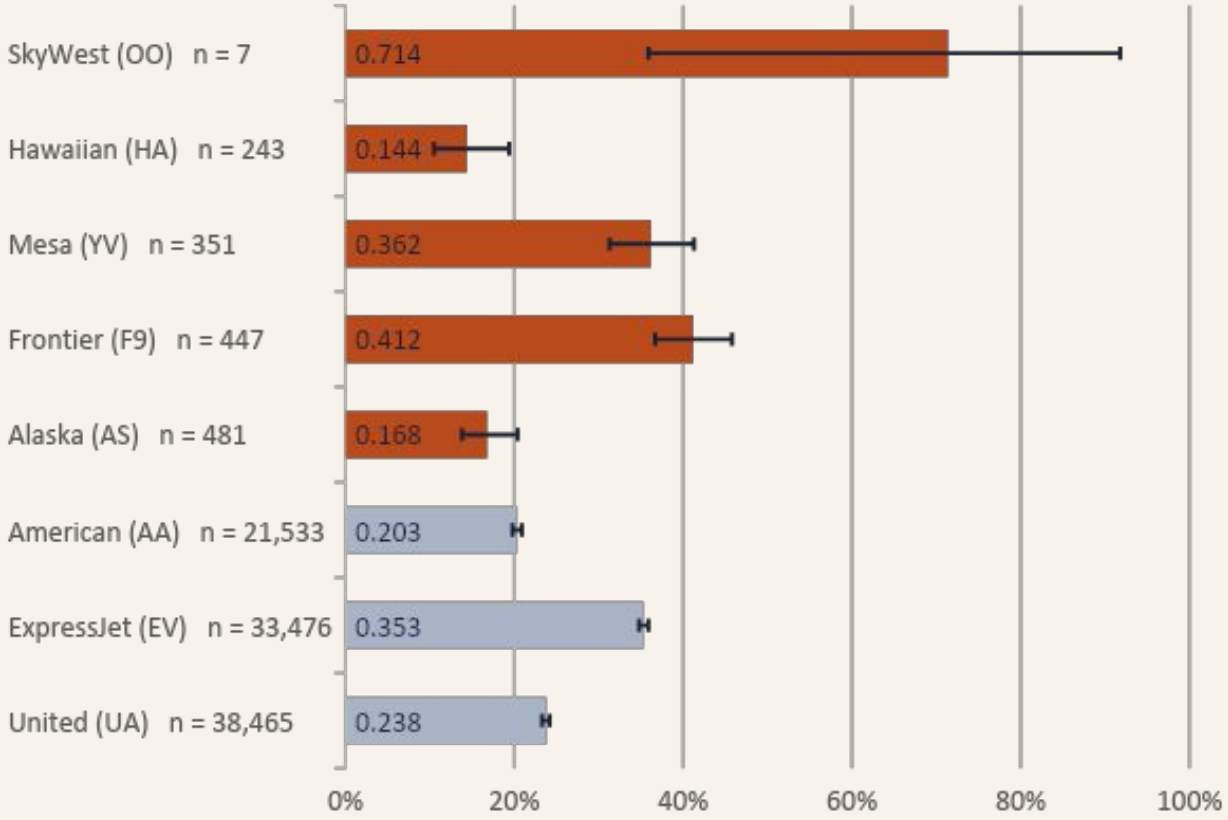
By origin airport



EDA on Jan–Aug only (D-101) so test months could not shape feature design; notebook 02. Rates are descriptive, not causal.

A delay rate is only as trustworthy as the number of flights behind it

Delay rate with 95% Wilson interval, by airline



Orange: fewer than 1,000 flights. Lines show the 95% interval.

THE WILSON INTERVAL (k delayed of n flights, p = k/n, z = 1.96)

$$\text{centre} = (p + z^2/2n) \div (1 + z^2/n)$$

$$\text{half-width} = z \cdot \sqrt{(p(1-p)/n + z^2/4n^2)} \div (1 + z^2/n)$$

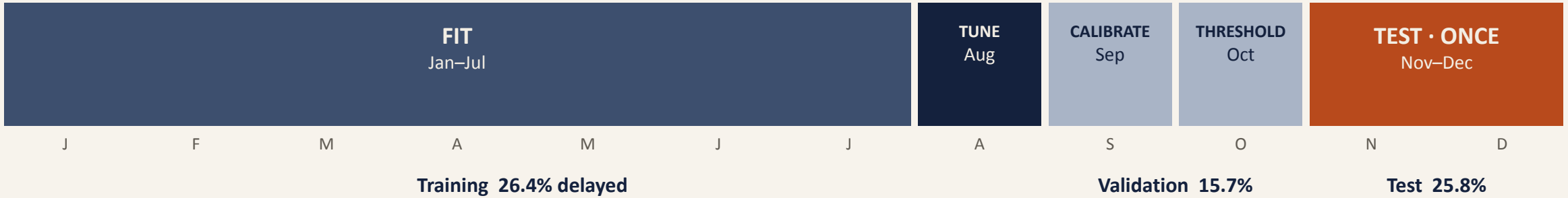
The extra z^2 terms pull small samples toward 50% and widen the range; for large n they vanish and it becomes the familiar $p \pm 1.96 \cdot \sqrt{(p(1-p)/n)}$.

Worked example: why not the simple formula?

Sample	Simple ± formula	Wilson
1 delayed of 10	−0.09 to 0.29 (impossible)	0.02 to 0.40
3 of 10	0.02 to 0.58	0.11 to 0.60
300 of 1,000	0.27 to 0.33	0.27 to 0.33

Airline rates and 95% Wilson intervals: notebook 02 (Jan–Aug), computed by rate_table / wilson_interval in src/flight_delay/data/validate.py.

Each period answers one question — and the test months were sealed before any analysis



THE TIMING RULE

A past flight counts only once its outcome is known: actual arrival in New York time + 30 minutes. It is usable from the next day.
Example: a Jan 10 JFK→LAX flight landing at 00:55 LA time (03:55 New York, Jan 11) can only inform flights from Jan 12 onward.

0

of 273,355 training and validation rows changed when every Nov–Dec record was removed

1×

the test set was evaluated — after the model, calibration and threshold were frozen

Encoder, scaler and imputer are fitted on training data only. Month and day-of-year are excluded: every later month is unseen in training.

Leakage policy: D-002, D-012, D-105; one-job-per-period protocol: D-201, D-203. Leakage framing [7]. Delay rates: notebooks 01–02 and D-205.

59 features, every one computable by the end of the previous day

Raw schedule
336,776 flights

↓

Seal Nov–Dec
Jan–Oct dev file

↓

Build features
schedule + past-only history

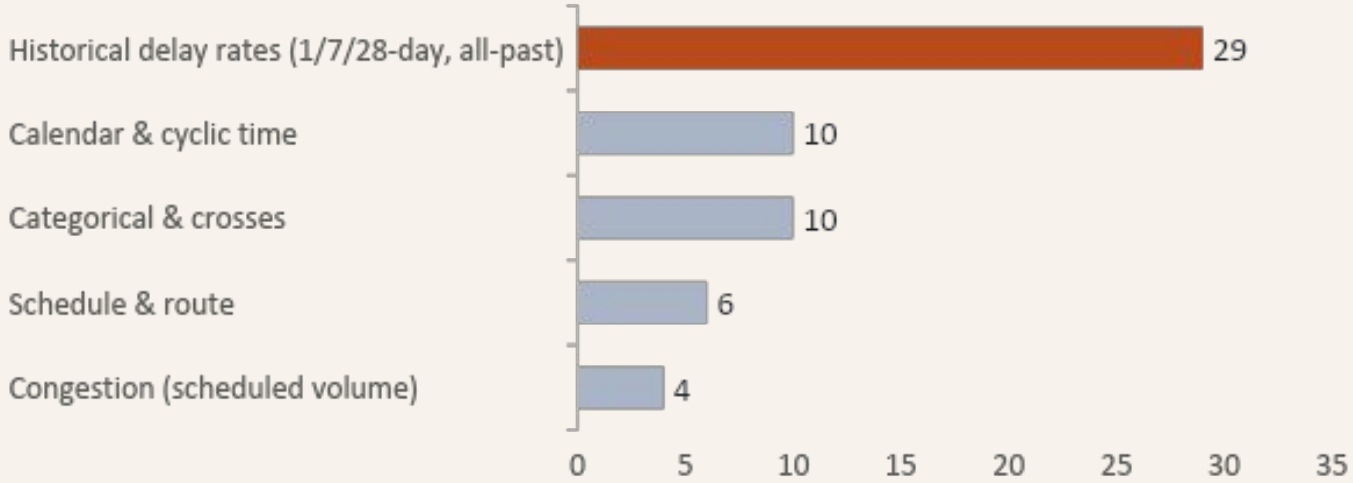
↓

Encode
fitted on training only

↓

Model input
49 numeric + 10 categorical

Features per group



Built in: sine/cosine hour and weekday (23:00 sits next to 00:00); scheduled block time on a common UTC basis; history rates shrunk toward the global past rate when a route has few flights. **Left out:** departure delay, actual times, tail number (leak the outcome); month and day-of-year (unseen months).

Feature set: D-102 to D-105, docs/feature_dictionary.md; time zones from airportsdata [3]; models via scikit-learn [8] and XGBoost [9].

Two very different model families gave almost the same answer

LOGISTIC REGRESSION

the linear baseline

$$p = \sigma(w \cdot x + b)$$

Weighted sum of features → sigmoid → probability

L2 penalty; C = 0.01 is strong shrinkage

class_weight = balanced: delayed flights count more, which inflates raw probabilities

0.401

PR-AUC · ROC-AUC 0.676 · selected

XGBOOST

gradient-boosted trees

$$F(x) = \Sigma\ \text{trees}(x)$$

Each new tree corrects the errors of the ones before

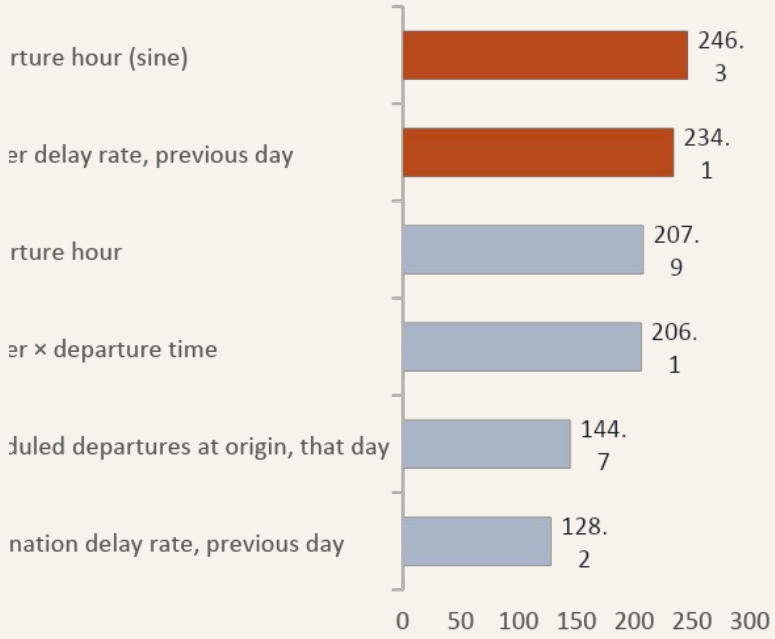
Depth 3 or 5 limits interactions; learning rate 0.03 or 0.1

Early stopping on August: best run used 16 trees

0.396

PR-AUC · ROC-AUC 0.677 · best of 4

What XGBoost leaned on (gain, top 6)



Time of day, carrier and yesterday's delays carry the signal — the same patterns the EDA found. Importance shows reliance, not cause.

Best run per family, fitted Jan–Jul, scored on August (D-202); importance: reports/results/d201_xgboost_gain.csv; models [8], [9].

The plan came first: 12 pre-registered runs, one rule, one winner retrained

1 Plan
grid and rule committed (bf7a51c)

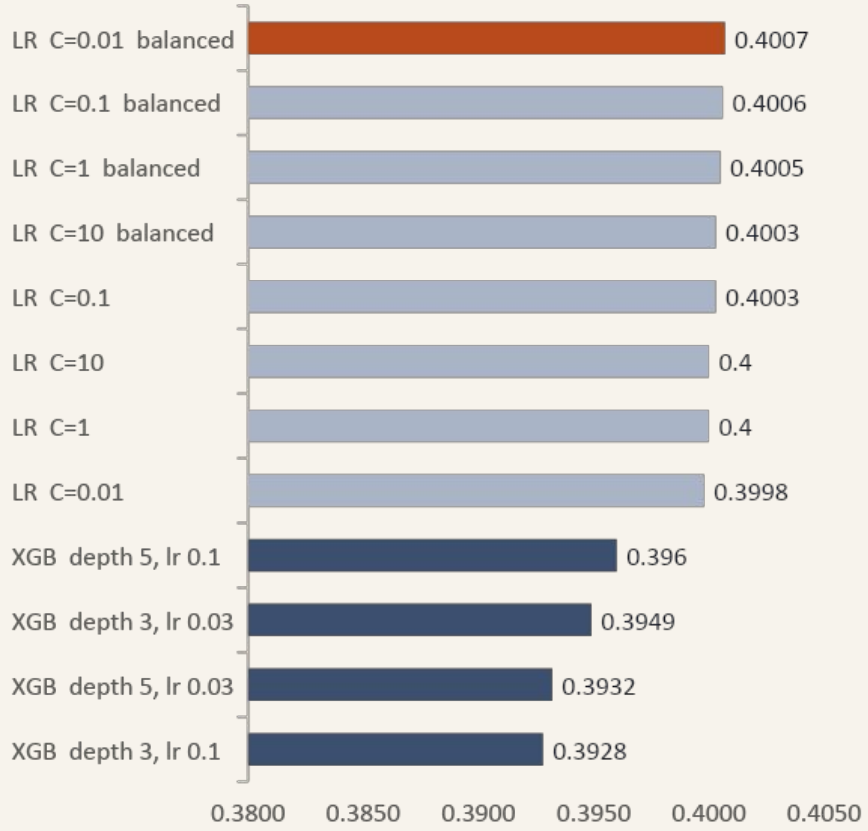
2 Fit
12 runs on Jan–Jul

3 Score
August PR-AUC; XGBoost stops early

4 Select
highest PR-AUC, ties → simpler

5 Retrain
winner on Jan–Aug, then frozen

August PR-AUC, all 12 runs (axis from 0.38)



HOW TO READ IT

Random ranking = 0.246 (August delay rate). Every run is about 1.6× better.

REPRODUCIBLE

Seed 42 · every run logged · scripts refuse to overwrite results.

LR reproduces exactly; XGBoost early stopping varies by library version (0.396 vs 0.393), without changing the winner.

Plan committed before training (D-201, bf7a51c); runs: D-202, models/d201/selection.json; PR-AUC [10]. August scores are tuning scores.

On unseen months the model ranks 1.5× better than random, and calibration cut ECE by 62%

0.398

PR-AUC on 53,991 test flights (random ranking = 0.258 delay rate)

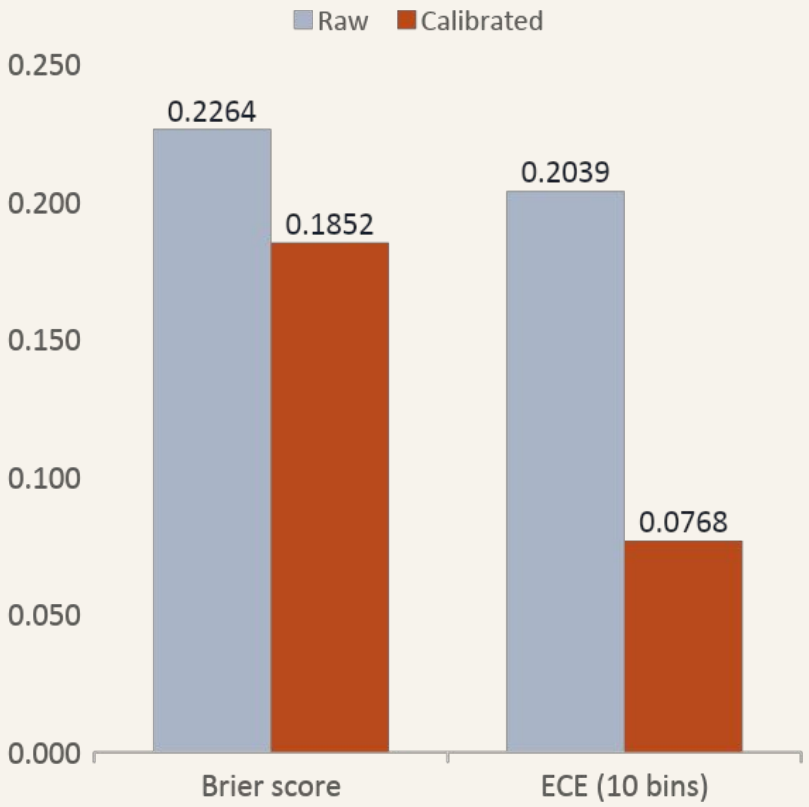
0.664

ROC-AUC — close to validation (0.67–0.69), so no collapse on new months

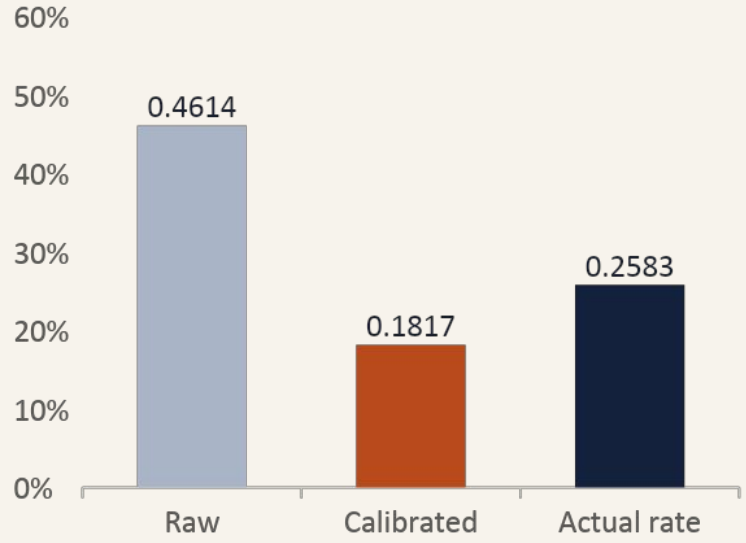
0.077

expected calibration error after sigmoid calibration (raw: 0.204)

Probability error (lower is better)



Average predicted vs actual

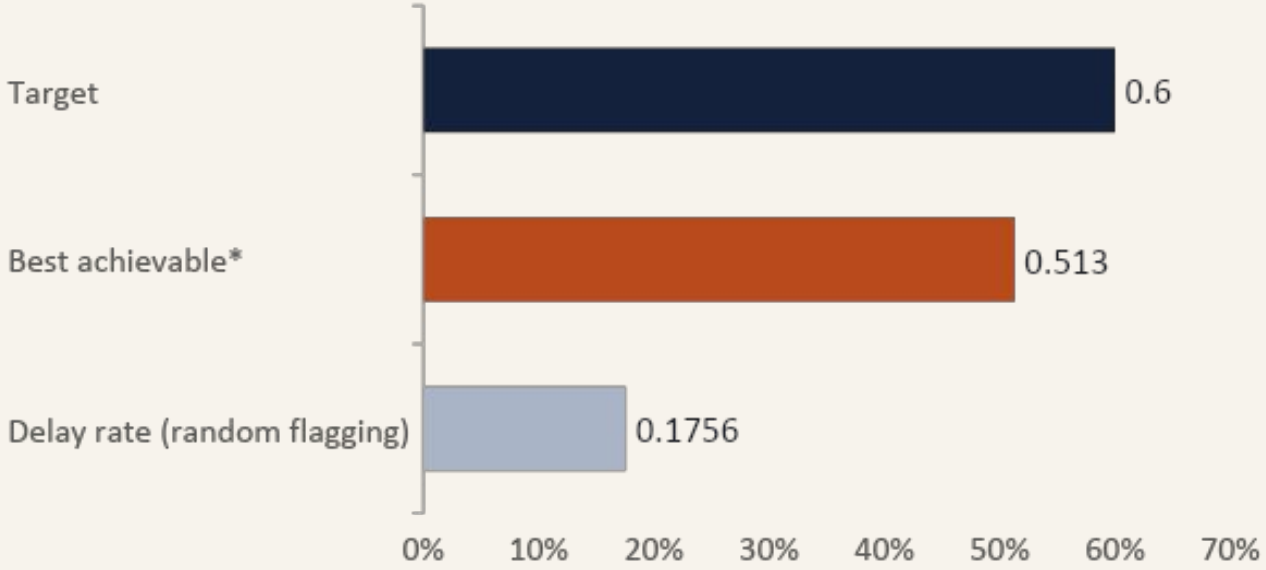


Calibrated probabilities run low (18% vs 26%): calibration was learned on September, the calmest month (13.7% delayed).

Single evaluation, frozen commit e559e20 (D-205); reports/results/test_metrics.json. ECE as in [12]; Brier score [13]; sigmoid calibration [11].

No threshold reached the 60% precision target — and the target was not lowered

Precision on October flights



* Best precision with at least 20 delayed flights flagged: 51.3%, flagging 158 flights and catching 1.6% of delayed ones. A diagnostic only — not a chosen operating point.

0.525

highest calibrated probability given to any October flight — the model is never confident enough for 60% precision

WHY THIS IS THE HONEST RESULT

Planners act on every flag, so false alarms cost money (D-012). Precision ≥ 60% was fixed before any result.

Lowering it after seeing 51% would tune the target to the data. Threshold metrics and the rescheduling rule are reported as not available.

Rule and target: D-015 (assumption), D-203; October results: D-204, models/d203/operating_point.json (28,618 October flights).

A day ahead, the schedule supports honest risk ranking — but not confident flight-level flags

WHAT HOLDS UP

Leakage-safe pipeline: timing rule, sealed test set, tests that pin it

Consistent ranking: PR-AUC 0.398 vs 0.258, ROC-AUC 0.664 on unseen months

Calibrated probabilities: ECE 0.204 → 0.077

LIMITATIONS

No weather or same-day data at decision time

Calm validation months: calibration under-predicts in Nov–Dec

60% precision unreachable; one year, three airports

30-minute reporting buffer is an assumption

NEXT STEPS

Add weather forecasts available at D−1

Rolling recalibration on recent months

Set the operating point from real cost data

Phase 2: temporal deep learning with learned embeddings

Disclosure (D-000): an earlier iteration evaluated on Nov–Dec; every model, feature and threshold here was re-decided without those results.

Sources used in the project

[1] M. Sajadi, “Flights” dataset, Kaggle, 2023. kaggle.com/datasets/matinsajadi/flights

[2] U.S. Bureau of Transportation Statistics, “Airline On-Time Performance Data,” TranStats. transtats.bts.gov/DatabaseInfo.asp?QO_VQ=EFD

[3] M. Borsetti, airportsdata v20250909, PyPI, 2025. pypi.org/project/airportsdata

[4] L. Carvalho et al., “On the relevance of data science for flight delay research: a systematic review,” Transport Reviews 41(4), 2021. doi:10.1080/01441647.2020.1861123

[5] J. J. Rebollo, H. Balakrishnan, “Characterization and prediction of air traffic delays,” Transp. Res. Part C 44, 2014. doi:10.1016/j.trc.2014.04.007

[6] S. Choi et al., “Prediction of weather-induced airline delays based on machine learning algorithms,” IEEE/AIAA DASC, 2016. doi:10.1109/DASC.2016.7777956

[7] S. Kaufman et al., “Leakage in data mining: formulation, detection, and avoidance,” ACM TKDD 6(4), 2012. doi:10.1145/2382577.2382579

[8] F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” JMLR 12, 2825–2830, 2011. jmlr.org/papers/v12/pedregosa11a.html

[9] T. Chen, C. Guestrin, “XGBoost: A Scalable Tree Boosting System,” KDD, 2016. doi:10.1145/2939672.2939785

[10] T. Saito, M. Rehmsmeier, “The precision-recall plot is more informative than the ROC plot … on imbalanced datasets,” PLoS ONE 10(3), 2015. doi:10.1371/journal.pone.0118432

[11] A. Niculescu-Mizil, R. Caruana, “Predicting good probabilities with supervised learning,” ICML, 2005. doi:10.1145/1102351.1102430

[12] C. Guo et al., “On calibration of modern neural networks,” ICML (PMLR 70), 2017. proceedings.mlr.press/v70/guo17a.html

[13] G. W. Brier, “Verification of forecasts expressed in terms of probability,” Monthly Weather Review 78(1), 1950. doi:10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2

Full bibliography with verification notes: report/references.bib and docs/literature_review.md.